

DBMS ASSIGNMENT – 2: ER DIAGRAM DESIGN AND RELATIONAL SCHEMA APPLICATION

Domain: Online Food Delivery System

- **Subject:** Database Management Systems
- **Assignment:** Assignment – 2
- **Topic:** ER Diagram Design and Relational Schema
- **Application Domain:** Online Food Delivery System

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1. INTRODUCTION

The Online Food Delivery System is a database-driven application designed to manage the process of ordering food from restaurants through an online platform. The system allows customers to maintain their personal information, save multiple delivery addresses, browse restaurant menu items, place orders, make payments, and receive food through delivery partners.

From the database perspective, the system needs to store information about customers, restaurants, menu items, orders, individual order items, delivery partners, addresses, and payments. The ER model provides a structured representation of these entities and the relationships between them. The relational schema then converts this conceptual ER model into a collection of tables that can be implemented in a relational database such as MySQL.

The provided Assignment 2 specification states that the ER model must retain all eight entities from the earlier Online Food Delivery System design and convert them into a matching relational schema.

2. OBJECTIVE

The main objective of this assignment is to design a complete Entity–Relationship (ER) Diagram for an Online Food Delivery System and convert that model into a Relational Schema.

The objectives are:

1. To identify the major entities involved in the application.
2. To identify the attributes of every entity.
3. To identify suitable primary keys.
4. To identify foreign-key relationships.
5. To determine the cardinality of each relationship.
6. To represent the system using a proper ER diagram.
7. To convert the ER model into relational tables.
8. To resolve the many-to-many relationship between Orders and Menu_Item using an associative entity.

9. To maintain data integrity through primary-key and foreign-key constraints.
10. To produce a database design suitable for implementation in MySQL.

3. APPLICATION OVERVIEW

3.1 Application Name

Online Food Delivery System

3.2 Purpose

An Online Food Delivery System enables customers to browse restaurants, view available menu items, place food orders, make payments, and receive their orders through delivery partners. The database stores and manages interactions between:

- Customers
- Restaurants
- Menu Items
- Orders
- Order Items
- Delivery Partners
- Payments
- Delivery Addresses

4. SCOPE OF THE SYSTEM

The database covers the following major activities:

- **Customer Management:** The system stores customer profile information such as name, email, phone number, and password.
- **Address Management:** Customers can save multiple delivery addresses such as home, work, or other locations.
- **Restaurant Management:** The system maintains restaurant information, contact details, physical address, and rating.

- **Menu Management:** Each restaurant can provide multiple food items with prices, descriptions, and availability information.
- **Order Management:** Customers can place orders for food from restaurants. Each order stores transaction information such as order date, status, and total amount.
- **Order Item Management:** Individual food items inside an order are represented through the Order_Item table.
- **Payment Management:** The database stores payment details associated with orders.
- **Delivery Management:** Delivery partners are assigned to orders and their information is maintained in the database.

5. ENTITIES AND ATTRIBUTES

5.1 CUSTOMER

Purpose: The Customer entity stores the account and contact information of users who place food orders.

Attribute	Data Type	Key	Description
Customer_ID	INT	PK ▾	Unique identifier for each customer
Name	VARCHAR(100)	— ▾	Full name of customer
Email	VARCHAR(150)	UNIQUE ▾	Login/contact email
Phone	VARCHAR(15)	UNIQUE ▾	Primary contact number

Attribute	Data Type	Key	Description
Password	VARCHAR(255)	— ▾	Hashed account password

- **Primary Key:** Customer_ID
- The Customer entity represents the main user of the application.

5.2 ADDRESS

Purpose: The Address entity stores the delivery locations saved by customers.

Attribute	Data Type	Key	Description
Address_ID	INT ▾	PK ▾	Unique identifier for an address
Customer_ID	INT ▾	FK ▾	Customer who owns the address
House_No	VARCHAR(30) ▾	— ▾	House/flat/apartment number
Street	VARCHAR(150) ▾	— ▾	Street name
City	VARCHAR(100) ▾	— ▾	City name
Pincode	VARCHAR(10) ▾	— ▾	Postal code

- **Primary Key:** Address_ID
- **Foreign Key:** Customer_ID → Customer.Customer_ID
- Each customer can maintain multiple addresses.

5.3 RESTAURANT

Purpose: The Restaurant entity stores information about food establishments participating in the platform.

Attribute	Data Type	Key	Description
Restaurant_ID	INT	PK ▾	Unique restaurant identifier
Restaurant_Name	VARCHAR(150)	— ▾	Name of restaurant
Phone	VARCHAR(15)	— ▾	Restaurant contact number
Address	VARCHAR(255)	— ▾	Physical restaurant location
Rating	DECIMAL(2,1)	— ▾	Average customer rating

- **Primary Key:** Restaurant_ID

5.4 MENU_ITEM

Purpose: The Menu_Item entity stores individual food products offered by restaurants.

Attribute	Data Type	Key	Description
Item_ID	INT ▾	PK ▾	Unique food-item identifier
Restaurant_ID	INT ▾	FK ▾	Restaurant offering the item
Item_Name	VARCHAR(150) ▾	— ▾	Name of food item
Description	VARCHAR(500) ▾	— ▾	Description of the item
Price	DECIMAL(10,2) ▾	— ▾	Current unit price
Availability	BOOLEAN ▾	— ▾	Whether the item is currently orderable

- **Primary Key:** Item_ID
- **Foreign Key:** Restaurant_ID → Restaurant.Restaurant_ID

5.5 DELIVERY_PARTNER

Purpose: The Delivery_Partner entity stores the information of personnel responsible for delivering customer orders.

Attribute	Data Type	Key	Description
Delivery_ID	INT	PK ▾	Unique delivery-partner identifier
Name	VARCHAR(100)	— ▾	Full name
Phone	VARCHAR(15)	UNIQUE ▾	Contact number
Vehicle_Number	VARCHAR(20)	UNIQUE ▾	Registered vehicle number
Status	VARCHAR(30)	— ▾	Current availability/status

- **Primary Key:** Delivery_ID

5.6 ORDERS

Purpose: The Orders entity stores the master information of every order placed by customers.

Attribute	Data Type	Key	Description
Order_ID	INT ▾	PK ▾	Unique order identifier
Customer_ID	INT ▾	FK ▾	Customer who placed the order

Attribute	Data Type	Key	Description
Restaurant_ID	INT ▾	FK ▾	Restaurant fulfilling the order
Address_ID	INT ▾	FK ▾	Delivery address
Delivery_ID	INT ▾	FK ▾	Assigned delivery partner
Order_Date	DATETIME ▾	— ▾	Date and time of order
Order_Status	VARCHAR(30) ▾	— ▾	Current order status
Total_Amount	DECIMAL(10,2) ▾	— ▾	Total value of order

- **Primary Key:** Order_ID
- **Foreign Keys:**
 - Customer_ID → Customer.Customer_ID
 - Restaurant_ID → Restaurant.Restaurant_ID
 - Address_ID → Address.Address_ID
 - Delivery_ID → Delivery_Partner.Delivery_ID
- **Example Order Statuses:** Placed → Preparing → Out for Delivery → Delivered (or Cancelled)

5.7 ORDER_ITEM

Purpose: The Order_Item entity stores the individual menu items included in an order. This is an associative entity used to resolve the many-to-many relationship between Orders and Menu_Item.

Attribute	Data Type	Key	Description
Order_ID	INT ▾	PK, FK ▾	Related order
Item_ID	INT ▾	PK, FK ▾	Ordered menu item
Quantity	INT ▾	— ▾	Number of units ordered

Attribute	Data Type	Key	Description
Unit_Price	DECIMAL(10,2) ▾	— ▾	Price at order time

- **Composite Primary Key:** (Order_ID, Item_ID)
- **Foreign Keys:**
 - Order_ID → Orders.Order_ID
 - Item_ID → Menu_Item.Item_ID

5.8 PAYMENT

Purpose: The Payment entity stores financial transaction information associated with an order.

Attribute	Data Type	Key	Description
Payment_ID	INT ▾	PK ▾	Unique payment identifier
Order_ID	INT ▾	FK, UNIQU... ▾	Order being paid for
Payment_Method	VARCHAR(30) ▾	— ▾	UPI / Card / Cash on Delivery
Payment_Status	VARCHAR(30) ▾	— ▾	Payment transaction status
Payment_Date	DATETIME ▾	— ▾	Date/time of payment
Amount	DECIMAL(10,2) ▾	— ▾	Amount charged

- **Primary Key:** Payment_ID
- **Foreign Key:** Order_ID → Orders.Order_ID (UNIQUE)

6. RELATIONSHIPS AND CARDINALITIES

Entity A	Relationship	Entity B	Cardinality
Customer ▾	Has ▾	Address ▾	1 : N ▾
Customer ▾	Places ▾	Orders ▾	1 : N ▾
Restaurant ▾	Offers ▾	Menu_Item ▾	1 : N ▾
Restaurant ▾	Fulfils ▾	Orders ▾	1 : N ▾
Delivery_Partner ▾	Handles ▾	Orders ▾	1 : N ▾
Address ▾	Used For ▾	Orders ▾	1 : N ▾
Orders ▾	Contains ▾	Menu_Item ▾	M : N ▾
Orders ▾	Has ▾	Payment ▾	1 : 1 ▾

7. ER DIAGRAM

Online Food Delivery System – Entity Relationship Diagram

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Figure Caption:

Figure 1: Entity–Relationship Diagram for the Online Food Delivery System

The ER diagram represents:

- All eight entities
- Attributes of each entity
- Primary keys
- Foreign keys
- Relationships
- Relationship names

- Cardinalities

8. EXPLANATION OF THE ER DIAGRAM

The ER diagram provides a visual representation of how the components of the Online Food Delivery System interact. At the center of the system is the **Orders** entity because an order connects multiple operational parts of the system: Customer → Orders → Restaurant → Menu Items → Order Items → Payment / Delivery

- A customer can place many orders. Every order belongs to one customer.
- A restaurant can receive many orders and can offer many menu items.
- The **Order_Item** entity plays an important role because an order can contain multiple menu items, and the same menu item can occur in multiple orders.
- The delivery process is represented by the relationship between **Delivery_Partner** and **Orders**.
- Payment information is stored separately in the **Payment** entity.

9. RELATIONAL SCHEMA

9.1 CUSTOMER

CUSTOMER (Customer_ID PK, Name, Email, Phone, Password)

9.2 ADDRESS

ADDRESS (Address_ID PK, Customer_ID FK → CUSTOMER.Customer_ID, House_No, Street, City, Pincode)

9.3 RESTAURANT

RESTAURANT (Restaurant_ID PK, Restaurant_Name, Phone, Address, Rating)

9.4 MENU_ITEM

MENU_ITEM (Item_ID PK, Restaurant_ID FK → RESTAURANT.Restaurant_ID, Item_Name, Description, Price, Availability)

9.5 DELIVERY_PARTNER

DELIVERY_PARTNER (Delivery_ID PK, Name, Phone, Vehicle_Number, Status)

9.6 ORDERS

ORDERS (Order_ID PK, Customer_ID FK → CUSTOMER.Customer_ID, Restaurant_ID FK → RESTAURANT.Restaurant_ID, Address_ID FK → ADDRESS.Address_ID, Delivery_ID FK → DELIVERY_PARTNER.Delivery_ID, Order_Date, Order_Status, Total_Amount)

9.7 ORDER_ITEM

ORDER_ITEM (Order_ID PK, FK → ORDERS.Order_ID, Item_ID PK, FK → MENU_ITEM.Item_ID, Quantity, Unit_Price)

9.8 PAYMENT

PAYMENT (Payment_ID PK, Order_ID FK → ORDERS.Order_ID UNIQUE, Payment_Method, Payment_Status, Payment_Date, Amount)

10. PRIMARY KEY IDENTIFICATION

Table	Primary Key
Customer	Customer_ID
Address	Address_ID
Restaurant	Restaurant_ID

Table	Primary Key
Menu_Item	Item_ID
Delivery_Partner	Delivery_ID
Orders	Order_ID
Order_Item	Order_ID + Item_ID
Payment	Payment_ID

11. FOREIGN KEY IDENTIFICATION

Table	Foreign Key	References
Address ▾	Customer_ID ▾	Customer.Customer_ID ▾
Menu_Item ▾	Restaurant_ID ▾	Restaurant.Restaurant... ▾
Orders ▾	Customer_ID ▾	Customer.Customer_ID ▾
Orders ▾	Restaurant_ID ▾	Restaurant.Restaurant... ▾
Orders ▾	Address_ID ▾	Address.Address_ID ▾
Orders ▾	Delivery_ID ▾	Delivery_Partner.Deliv... ▾
Order_Item ▾	Order_ID ▾	Orders.Order_ID ▾
Order_Item ▾	Item_ID ▾	Menu_Item.Item_ID ▾
Payment ▾	Order_ID ▾	Orders.Order_ID ▾

12. ASSOCIATIVE ENTITY – ORDER_ITEM

One of the most important database design elements in this application is the **Order_Item** table.

Many-to-Many Relationship

There is an M:N relationship between:

Orders ↔ Menu_Item

This means:

- One order can contain many menu items.
- One menu item can appear in many different orders.

A direct M:N relationship is not represented as a simple foreign key in a relational table. Therefore, the relationship is resolved using:

ORDER_ITEM (Order_ID, Item_ID, Quantity, Unit_Price)

The composite key (**Order_ID**, **Item_ID**) uniquely identifies each item within a particular order. The table additionally stores Quantity and the historical Unit_Price at the time the order was placed.

13. DETAILED RELATIONSHIP EXPLANATION

13.1 Customer – Address

- **Cardinality:** 1:N
- One customer can save multiple delivery addresses (e.g., Home, College, Office). Each address record belongs to exactly one customer.

13.2 Customer – Orders

- **Cardinality:** 1:N
- One customer can place many orders over time. Each individual order belongs to one customer.

13.3 Restaurant – Menu_Item

- **Cardinality:** 1:N
- A restaurant can offer many menu items. Each menu item belongs to one restaurant.

13.4 Restaurant – Orders

- **Cardinality:** 1:N
- A restaurant can receive and fulfill many orders. Each order is associated with one restaurant in the designed schema.

13.5 Delivery_Partner – Orders

- **Cardinality:** 1:N
- A delivery partner can handle multiple orders over time. Each order is assigned to one delivery partner in the specified model.

13.6 Address – Orders

- **Cardinality:** 1:N
- A saved address may be used for multiple orders. Each order specifies one delivery address.

13.7 Orders – Menu_Item

- **Cardinality:** M:N
- An order can contain several menu items, and a menu item can appear in many different orders. Resolved by `ORDER_ITEM`.

13.8 Orders – Payment

- **Cardinality:** 1:1
- Each order has exactly one payment record, and each payment belongs to exactly one order. Enforced using `Payment.Order_ID UNIQUE`.

14. DATABASE INTEGRITY AND CONSTRAINTS

14.1 Primary Key Constraint

Primary keys ensure that every record can be uniquely identified.

14.2 Foreign Key Constraint

Foreign keys connect related entities and help maintain referential integrity.

14.3 UNIQUE Constraint

Unique constraints prevent duplicate values in selected attributes:

- `Customer.Email`
- `Customer.Phone`
- `Delivery_Partner.Phone`
- `Delivery_Partner.Vehicle_Number`
- `Payment.Order_ID`

14.4 Composite Primary Key

`Order_Item` contains (`Order_ID` + `Item_ID`) as its composite primary key to ensure uniqueness per order-item pairing.

15. DESIGN AND NORMALIZATION CONSIDERATIONS

The database is organized into separate relations for different functional areas to eliminate redundancy and maintain integrity across all eight tables. Separating order-level information from item-level information allows an order to contain multiple items without creating repeating columns.

16. SYSTEM WORKFLOW

CUSTOMER

↓ Selects Restaurant

Views MENU_ITEM

↓ Creates ORDER

ORDER contains ORDER_ITEM

↓ Payment is recorded

Delivery Partner is assigned

↓ Order Status is updated

Food is delivered

Step-by-step process:

- **Step 1 – Customer Registration:** Stored in **Customer**.
- **Step 2 – Address Management:** Stored in **Address**.
- **Step 3 – Restaurant Selection:** Interacts with **Restaurant**.
- **Step 4 – Menu Browsing:** Retrieved from **Menu_Item**.
- **Step 5 – Order Creation:** Created in **Orders**.
- **Step 6 – Order Item Creation:** Items and quantities stored in **Order_Item**.
- **Step 7 – Payment:** Payment recorded in **Payment**.
- **Step 8 – Delivery:** Partner from **Delivery_Partner** associated with order.
- **Step 9 – Order Fulfillment:** Status progresses: **Placed** → **Preparing** → **Out for Delivery** → **Delivered**.

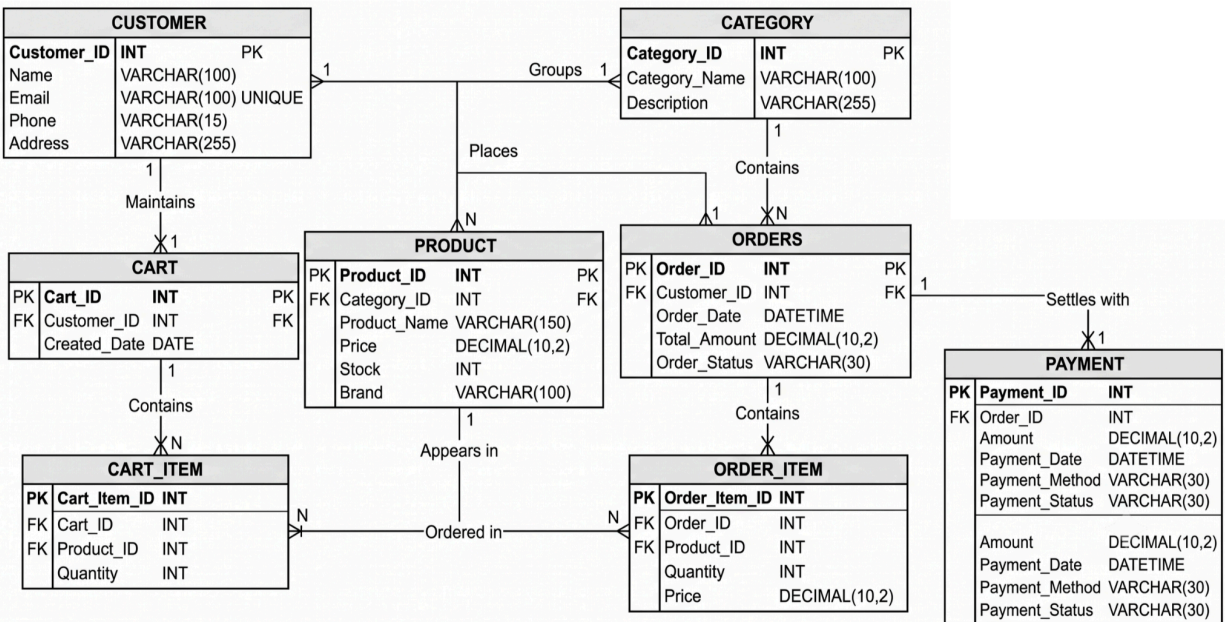
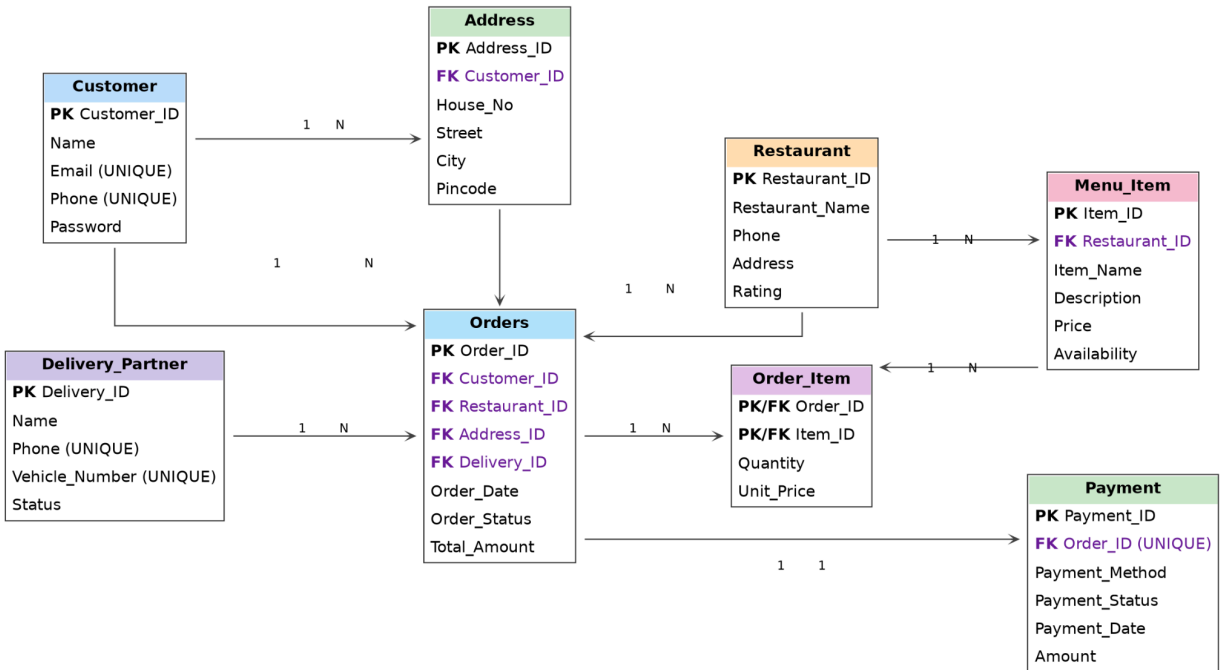
17. SCHEMA SUMMARY

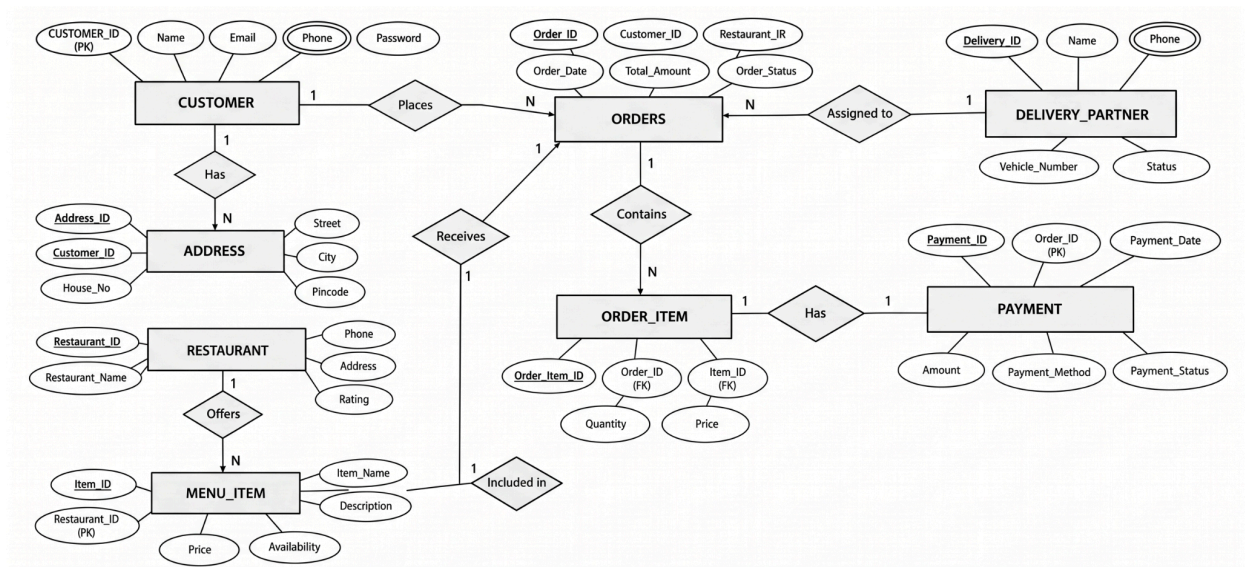
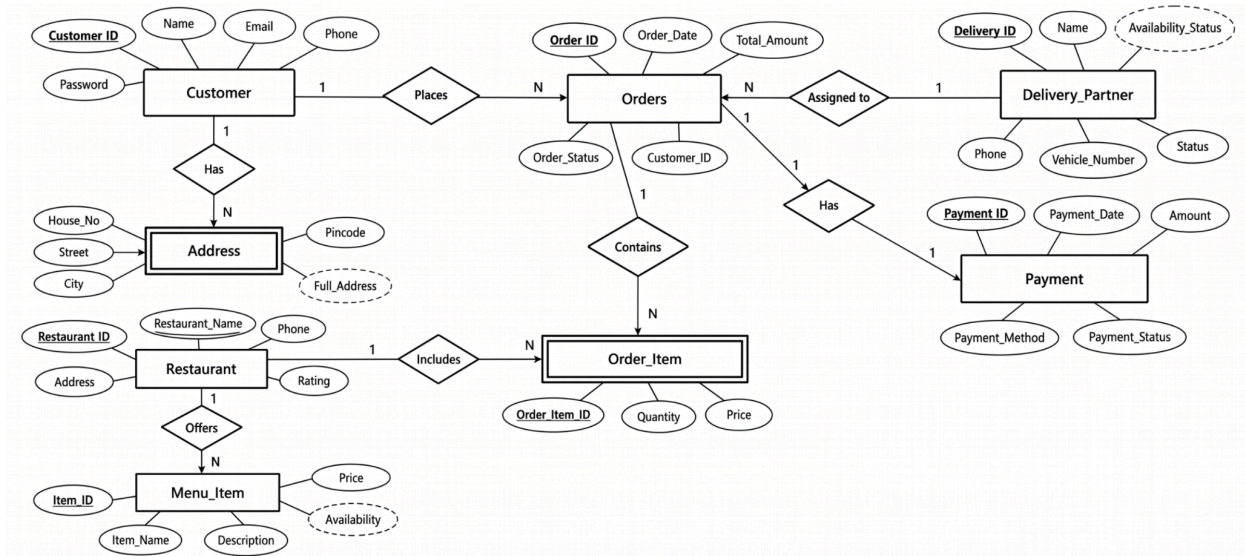
No.	Table	Main Purpose
1	CUSTOMER	Customer account information

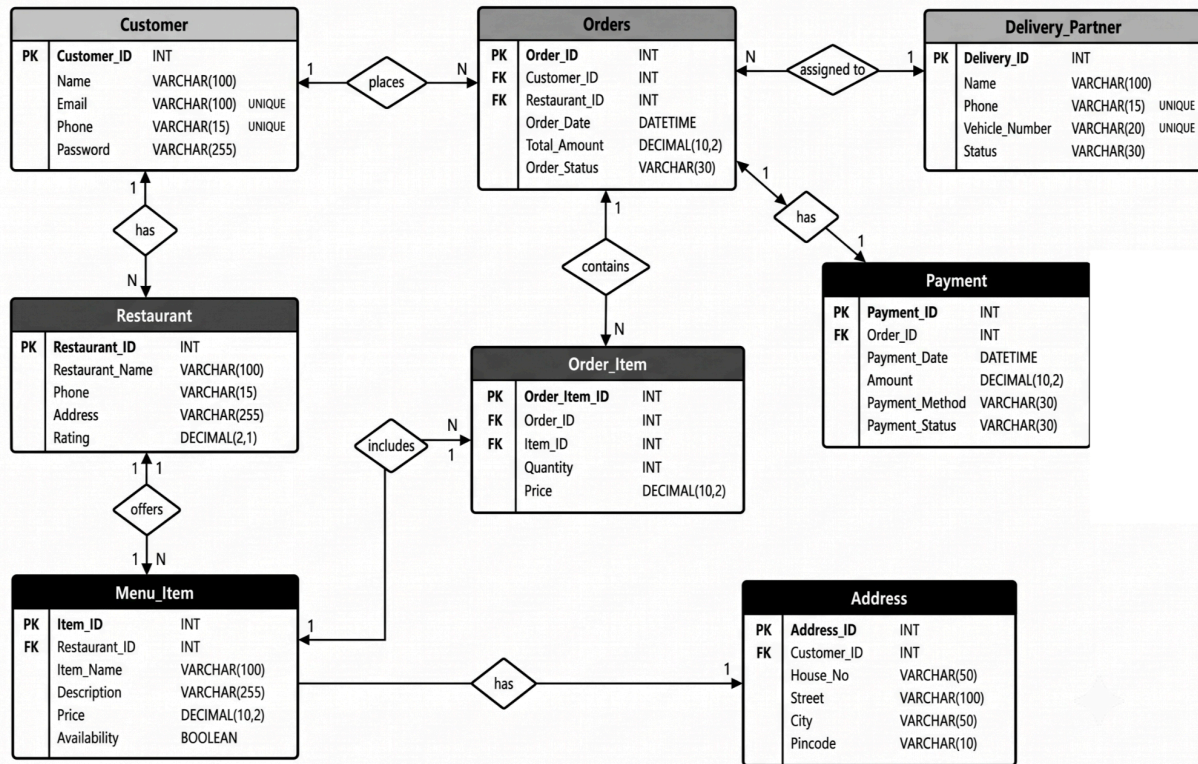
No.	Table	Main Purpose
2	ADDRESS	Customer delivery addresses
3	RESTAURANT	Restaurant information
4	MENU_ITEM	Food/menu catalog
5	ORDERS	Main order transaction
6	ORDER_ITEM	Individual items in each order
7	DELIVERY_PARTNER	Delivery personnel information
8	PAYMENT	Payment and transaction information

18. CONCLUSION

The Online Food Delivery System has been successfully modeled using an Entity–Relationship approach consisting of eight entities: Customer, Address, Restaurant, Menu_Item, Orders, Order_Item, Delivery_Partner, and Payment. The design resolves the M:N relationship between Orders and Menu_Item using the Order_Item associative table, enforces a 1:1 relationship between Orders and Payment via a UNIQUE constraint, and establishes key relationships and constraints suitable for implementation in MySQL.







ONLINE FOOD Delivery System Entity-Relationship (ER) DIAGRAM

